



Pancreatic Neuroendocrine Tumors (Islet Cell Tumors) Treatment (PDQ®)–Health Professional Version

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General Information About Pancreatic Neuroendocrine Tumors (Islet Cell Tumors)

Incidence and Mortality

Pancreatic neuroendocrine tumors (NETs) are uncommon cancers with about 1,000 new cases per year in the United States.[1] They account for less than 2% of pancreatic malignancies and, overall, have a better prognosis than the more common pancreatic exocrine tumors.[1,2]

Pathogenesis

Tumors of the endocrine pancreas are a collection of tumor cell types collectively referred to as pancreatic NETs. These tumors originate in islet cells. Although they may be similar or identical in histological appearance to carcinoid tumors of the gastrointestinal tract, differences in their underlying biology and likely differences in response to therapeutic agents suggest that they should be treated and investigated as a distinct entity.[3]

Most pancreatic NETs are sporadic, but some occur as part of the autosomal dominant multiple endocrine neoplasia type 1 (MEN1) inherited syndrome. This syndrome consists of tumors of the anterior pituitary, parathyroid, and endocrine pancreas glands and is a result of inactivation of the *MEN1* tumor suppressor gene located on chromosome 11q13.[4] When part of the MEN1 syndrome, there may be multiple pancreatic tumors.

Islet tumors may either be functional (produce one or more active hormones) or nonfunctional.[4] The functional tumors, which usually present with symptoms of hormone hypersecretion, include:

- Gastrinoma.
- Insulinoma.
- Glucagonoma.

- Somatostatinoma.
- VIPoma (Verner-Morrison syndrome).

Prognostic Factors

Most islet cell cancers are functional, but about 15% are nonfunctional, with presentations similar to the far more common exocrine adenocarcinomas of the pancreas.[5-7] Because of the presence of several cell types in the pancreatic islets (alpha, beta, delta, A, B, C, D, E, F), the term islet cell tumors refers to at least five distinct cancers that, when functional, produce unique metabolic and clinical characteristics. The clinical manifestations in functional tumors may result from the distinctive metabolic effects of the polypeptide(s) secreted by the cancer cells rather than from tumor bulk or metastatic disease. Functional tumors may even be too small to be detected by conventional imaging techniques.

Nonfunctional tumors tend to present at later clinical stages with symptoms attributable to mass effect or metastases.[4] Although nonfunctional tumors do not produce specific clinical syndromes, they may secrete inactive amine and peptide products such as:

- Neurotensin.
- Alpha-subunit of human chorionic gonadotropin (alpha-hCG).
- Neuron-specific enolase.
- Pancreatic polypeptide.
- Chromogranin A.

Diagnostics

The potential long delays between initial symptoms and diagnosis and the varied effects of the polypeptides secreted often necessitate involvement of multiple surgical and medical subspecialties. Surgery is the only curative modality and is often used, even in patients with metastatic disease, to alleviate the symptoms of hormonal hypersecretion.[4] Effective palliation may be achieved because of the slow-growing nature of most of these tumors and the potential use of antihormonal pharmacological therapy (e.g., cimetidine in the ulcer-producing Zollinger-Ellison syndrome). In patients with indolent, slow-growing, metastatic islet cell tumors, the best therapy may be careful observation, and no treatment until palliation is required. In patients with MEN1 in which 85% have pancreatic islet cell tumors, 90% have hyperparathyroidism, and 65% have pituitary tumors, and they are less likely to be cured by pancreatic resection than are patients with sporadic islet cell tumors. With the exception of pain relief from bone metastases, radiation therapy has a limited role in this disease.

Tumor localization and staging studies include imaging with computed tomography (CT) with or without magnetic resonance imaging (MRI), and endoscopic ultrasonography. Somatostatin-

receptor scintigraphy and single-photon emission CT may be useful adjuncts. However, somatostatin-receptor scintigraphy is less useful in localizing insulinomas versus other pancreatic NETs, since insulinomas often have a low density of somatostatin receptors.[4] If the noninvasive tests do not reveal a tumor, but clinical suspicion remains high, more invasive and technically demanding tests, such as selective arteriography or selective arterial stimulation (with a secretagogue specific for the suspected tumor type), may be useful.[7]

Some of the tumor types have unique characteristics that require specific approaches in their diagnosis and initial evaluation.

Gastrinoma

Diagnosis is dependent on elevated serum gastrin and elevated gastric acid levels. Provocative testing with calcium and secretin shows considerable overlap, and the value of these tests is limited. Zollinger-Ellison syndrome is a condition of unrelenting peptic ulcer disease, diarrhea, and gastric hyperacidity, associated with a gastrin-producing tumor. It accounts for less than 1% of all peptic ulcer disease. About 15% to 35% of gastrinomas are associated with MEN1 syndrome and up to 50% are malignant. Up to 33% of patients with gastrinomas have liver metastases.[4] For more information, see the [Diarrhea](#) section in Gastrointestinal Complications.

Diagnostic tests:

1. BAO:MAO ≥ 0.6 (basal acid output:maximal acid output).
2. Overnight AO ≥ 100 mmol.
3. BAO ≥ 10 mmol/hr.
4. Serum gastrin 10x normal or >500 pg/mL (the accuracy of gastrin assays may vary widely).
5. Secretin test: 1 unit/kg intravenous rapid injection: Positive = 100% increase in gastrin within 10 minutes; 2 units/kg: Positive = 100% increase over baseline.
6. Elevated hCG levels.

Insulinoma

Insulinomas are far more likely to be benign than malignant. Only 10% of insulinomas are multiple, and only 10% are malignant. About 5% to 8% are associated with MEN1 syndrome.[4] The clinical manifestations are those of hypoglycemia, which results from inappropriate secretion of insulin by the tumor. Fasting hypoglycemia (<40 mg/dL) associated with an elevated insulin level (in the absence of exogenous administration of insulin) is pathognomonic. Measurement of plasma proinsulin may be helpful for diagnosing insulin-secreting carcinomas. These tumors are usually slow-growing tumors and, when localized to the pancreas or regional lymph nodes, can be cured with pancreatic resection.

The approach to management depends on carefully performed preoperative localization studies and the findings at exploratory laparotomy. In a retrospective case series of 30 patients with 32 pancreatic insulinomas, the combination of preoperative, dual-phase, thin-section multidetector CT and endoscopic sonography correctly identified and localized all of the tumors.[8] These tests, with or without MRI, have replaced older, more invasive, and technically challenging tests, such as percutaneous transhepatic portal venous sampling and arterial stimulation with venous sampling, except for unusual circumstances in which the imaging tests are unsuccessful.[4,9]

Glucagonoma

Glucagonoma is the third most common endocrine-secreting islet cell tumor. About 75% of glucagonomas are malignant.[4] Necrolytic migratory erythema, hyperglycemia, and venous thrombosis comprise a virtually diagnostic triad. A serum glucagon level greater than 1,000 pg/mL confirms the diagnosis. These tumors tend to be large and easily visible on CT scan. Somatostatin-receptor scintigraphy scanning may be a useful adjunct in detecting metastases.

Miscellaneous islet cell tumors

These tumors are rare but have defined clinical syndromes associated with specific production of polypeptide hormone production by islet cell tumors. Because of their rarity and similar approaches to management, they are grouped in the section on treatment. Miscellaneous tumors include:

- VIPoma.

VIPoma is characterized by watery diarrhea, hypokalemia, and achlorhydria. A serum vasoactive intestinal peptide (VIP) greater than 200 pg/mL is diagnostic.[4] These tumors can generally be easily localized by CT scan. Somatostatin-receptor scintigraphy scanning may be a useful adjunct in detecting metastases.

- Somatostatinoma.

These tumors are particularly rare. They often present with diarrhea, steatorrhea, diabetes, and/or gallstones. Decreased pancreatic secretion of enzymes and bicarbonate accounts for the diarrhea and steatorrhea. Somatostatin-mediated inhibition of cholecystokinin leads to gallstone formation. Somatostatin also inhibits insulin, producing hyperglycemia. The diagnosis is made by a fasting serum somatostatin level greater than 100 pg/mL. CT scan, MRI, and endoscopic ultrasound can usually help localize and stage the tumor. Most of these tumors are malignant and have metastases at diagnosis.

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Cellular Classification of Pancreatic Neuroendocrine Tumors (Islet Cell Tumors)

Table 1. Endocrine Tumors of the Pancreas

Islet Cells	Secreted Active Agent	Tumor and Syndrome
Alpha	Glucagon	Glucagonoma (diabetes, dermatitis)
Beta	Insulin	Insulinoma (hypoglycemia)
Delta	Somatostatin	Somatostatinoma (mild diabetes); diarrhea/steatorrhea; gallstones
D	Gastrin	Gastrinoma (peptic ulcer disease)
A→D	VIP and/or other undefined mediators	WDHA
	5-HT	Carcinoid
	ACTH	Cushing disease
	MSH	Hyperpigmentation
Interacinar Cells	Secreted Active Agent	Tumor and Syndrome
F	Pancreatic polypeptide	Multiple hormonal syndromes
EC	5-HT	Carcinoid
5-HT = serotonin; ACTH = adrenocorticotropin hormone; MSH = melanocyte-stimulating hormone; VIP = vasoactive intestinal peptide; WDHA = watery diarrhea, hypokalemia, and achlorhydria.		

Stage Information for Pancreatic Neuroendocrine Tumors (Islet Cell Tumors)

American Joint Committee on Cancer (AJCC) Stage Groupings and TNM Definitions

The AJCC has designated staging by TNM (tumor, node, metastasis) classification to define pancreatic neuroendocrine tumors (islet cell tumors).^[1]

Table 2. Definitions for TNM Stage I^a

Stage	TNM	Definition
I	T1, N0, M0	T1 = Tumor limited to the pancreas, ^b <2 cm.
		N0 = No regional lymph node involvement.
		M0 = No distant metastasis.
T = primary tumor; N = regional lymph node; M = distant metastasis.		
^a Reprinted with permission from AJCC: Neuroendocrine Tumors of the Pancreas. In: Amin MB, Edge SB, Greene FL, et al., eds.: <i>AJCC Cancer Staging Manual</i> . 8th ed. New York, NY: Springer, 2017, pp. 407–19.		
^b The explanation for superscript b is at the end of Table 5 .		

Table 3. Definitions for TNM Stage II^a

Stage	TNM	Definition
II	T2, N0, M0	T2 = Tumqr limited to the pancreas, 2–4 cm.
		N0 = No regional lymph node involvement.
		M0 = No distant metastasis.
	T3, N0, M0	T3 = Tumqr limited to the pancreas, >4 cm; <i>or</i> tumor invading the duodenum or common bile duct.
		N0 = No regional lymph node involvement.
		M0 = No distant metastasis.
T = primary tumor; N = regional lymph node; M = distant metastasis.		
^a Reprinted with permission from AJCC: Neuroendocrine Tumors of the Pancreas. In: Amin MB, Edge SB, Greene FL, et al., eds.: <i>AJCC Cancer Staging Manual</i> . 8th ed. New York, NY: Springer, 2017, pp. 407–19.		
^b The explanation for superscript b is at the end of Table 5 .		

Table 4. Definitions for TNM Stage III^a

Stage	TNM	Definition
III	T4, N0, M0	T4 = Tumor invading adjacent organs (stomach, spleen, colon, adrenal gland) or the wall of large vessels (celiac axis or the superior mesenteric artery).
		N0 = No regional lymph node involvement.
		M0 = No distant metastasis.
	Any T, N1, M0	TX = Tumor cannot be assessed.
		T1 = Tumor limited to the pancreas, ^b <2 cm.
		T2 = Tumor limited to the pancreas, ^b 2–4 cm.
		T3 = Tumor limited to the pancreas, ^b >4 cm; or tumor invading the duodenum or common bile duct.
		T4 = Tumor invading adjacent organs (stomach, spleen, colon, adrenal gland) or the wall of large vessels (celiac axis or the superior mesenteric artery).
		N1 = Regional lymph node involvement.
		M0 = No distant metastasis.
T = primary tumor; N = regional lymph node; M = distant metastasis.		
^a Reprinted with permission from AJCC: Neuroendocrine Tumors of the Pancreas. In: Amin MB, Edge SB, Greene FL, et al., eds.: <i>AJCC Cancer Staging Manual</i> . 8th ed. New York, NY: Springer, 2017, pp. 407–19.		
^b The explanation for superscript b is at the end of Table 5 .		

Table 5. Definitions for TNM Stage IV^a

Stage	TNM	Definition
IV	Any T, Any N, M1	TX = Tumor cannot be assessed.
		T1 = Tumor limited to the pancreas, ^b <2 cm.
		T2 = Tumor limited to the pancreas, ^b 2–4 cm.
		T3 = Tumor limited to the pancreas, ^b >4 cm; <i>or</i> tumor invading the duodenum or common bile duct.
		T4 = Tumor invading adjacent organs (stomach, spleen, colon, adrenal gland) or the wall of large vessels (celiac axis or the superior mesenteric artery).
		NX = Regional lymph nodes cannot be assessed.
		N0 = No regional lymph node involvement.
		N1 = Regional lymph node involvement.
		M1 = Distant metastases.
		–M1a = Metastasis confined to liver.
		–M1b = Metastases in at least one extrahepatic site (e.g., lung, ovary, nonregional lymph node, peritoneum, bone).
		–M1c = Both hepatic and extrahepatic metastases.

Stage	TNM	Definition
T = primary tumor; N = regional lymph node; M = distant metastasis.		
^a Reprinted with permission from AJCC: Neuroendocrine Tumors of the Pancreas. In: Amin MB, Edge SB, Greene FL, et al., eds.: <i>AJCC Cancer Staging Manual</i> . 8th ed. New York, NY: Springer, 2017, pp. 407–19.		
^b <i>Limited to the pancreas</i> means there is no invasion of adjacent organs (stomach, spleen, colon, adrenal gland) or the wall of large vessels (celiac axis or the superior mesenteric artery). Extension of tumor into peripancreatic adipose tissue is NOT a basis for staging. <i>Note:</i> Multiple tumors should be designated as such (the largest tumor should be used to assign T category): if the number of tumors is known, use T(#); e.g., pT3(4) N0 M0; if the number of tumors is unavailable or too numerous, use the <i>m</i> suffix, T(m); e.g., pT3(m) N0 M0.		

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Treatment Option Overview for Pancreatic Neuroendocrine Tumors (Islet Cell Tumors)

Localized Disease

If technically and medically feasible, primary management of endocrine tumors of the pancreas involves surgical resection with curative intent. Given the rare nature of these tumors, surgical approaches are based on case series and expert opinion rather than randomized controlled trials.[1] The surgical options listed below are based on retrospective series from single reporting centers.[2-4][[Level of evidence C2](#)]

Adjuvant therapy has no proven benefit and is, therefore, investigational. There have been no well-controlled trials of adjuvant therapy after complete tumor resection.[5]

Surgical Cytoreduction for Metastases

Surgery plays a role even in the setting of metastatic disease. The symptoms of metastatic functional pancreatic neuroendocrine tumors (NETs) may be ameliorated by the reduction of overall tumor burden through surgical debulking.

The liver is a common site of metastasis from pancreatic NETs. Because of the slow growth rate of many NETs, liver metastases are often resected when technically feasible. Resection of all

grossly visible liver metastases can be associated with long-term survival and, in the case of symptomatic hormonally functional tumors, symptom relief.[6] Most symptoms from functional tumors respond to this form of surgical debulking. It is not known if the favorable survival rates are attributable to patient selection factors (e.g., underlying patient condition, extent of metastases, slow doubling time, and so forth).

Alternative approaches to the management of liver metastases have been reported, including Gelfoam embolization or transarterial chemoembolization,[7] radioembolization with radioactive microspheres,[8-10] radiofrequency ablation, cryoablation, and percutaneous alcohol ablation. These alternative approaches have been reviewed.[11]

Results from surgical resection series appear to be more favorable than with these techniques, and surgery is considered to be the standard approach to resectable liver metastases. However, there are no high-quality studies comparing the various approaches. A systematic review of evidence comparing liver resection versus other treatments for patients with resectable liver neuroendocrine metastases found no randomized trials, or even quasi-experimental, cohort, or case-control studies in which the patient population given the alternative therapies was similar enough to the surgery group to draw reliable conclusions.[12] The evidence for resection of all grossly visible liver metastases is derived solely from case series.[[Level of evidence C2](#)]

In most cases, liver metastases are not completely resectable. Cytoreductive surgery, with or without radiofrequency ablation or cryoablation, has been used to palliate symptoms. A systematic review found no randomized or quasi-randomized trials comparing cytoreductive surgery to other palliative approaches such as chemotherapy or tumor product inhibitors.[13] The evidence for surgical cytoreduction of unresectable liver metastases is restricted to case series [[Level of evidence C2](#)], and interpretation of outcomes may be strongly affected by patient selection factors.

Systemic Therapy for Advanced and Metastatic Disease

Somatostatin analogues may be effective in reducing the symptoms of functional tumors.[14]

Chemotherapy using drugs such as the following, either alone or in combination, has been shown to have antitumor effects, but evidence is weak or conflicting regarding the impact of chemotherapy on overall survival:[15-17]

- Streptozocin.
- Doxorubicin.
- Fluorouracil.
- Chlorozotocin.

- Dacarbazine.
- Temozolomide.

A variety of systemic agents have shown biological or palliative activity, including:[5,18]

- Tyrosine kinase inhibitors (e.g., sunitinib).
- Temozolomide.
- Vascular endothelial growth factor pathway inhibitors.
- Mammalian target of rapamycin inhibitors (e.g., everolimus).

Nearly all of the evidence of activity is derived from case series.[Level of evidence C3] However, there are ongoing placebo-controlled randomized trials of everolimus [19] and sunitinib [20] that have been reported in abstract form showing an increase in progression-free survival in each case.[Level of evidence B1]

Favorable responses have been reported in patients with advanced progressive pancreatic NETs after treatment with several radiolabeled somatostatin analogues in which the analogues octreotide, octreotate, lanreotide, or edotreotide are stably attached to the radionuclides indium In 111, yttrium Y 90, or lutetium Lu 177.[21-23] The relative efficacy of these various compounds is unknown. Study designs have been limited to case series with tumor response, biochemical response, or symptom control as the measure of efficacy.[Level of evidence C3]

As noted in each of the clinical situations, there is a paucity or absence of high-level evidence and a need for randomized controlled trials.[5]

Fluorouracil dosing

The *DPYD* gene encodes an enzyme that catabolizes pyrimidines and fluoropyrimidines, like capecitabine and fluorouracil. An estimated 1% to 2% of the population has germline pathogenic variants in *DPYD*, which lead to reduced DPD protein function and an accumulation of pyrimidines and fluoropyrimidines in the body.[24,25] Patients with the *DPYD**2A variant who receive fluoropyrimidines may experience severe, life-threatening toxicities that are sometimes fatal. Many other *DPYD* variants have been identified, with a range of clinical effects.[24-26] Fluoropyrimidine avoidance or a dose reduction of 50% may be recommended based on the patient's *DPYD* genotype and number of functioning *DPYD* alleles.[27-29] *DPYD* genetic testing costs less than \$200, but insurance coverage varies due to a lack of national guidelines.[30] In addition, testing may delay therapy by 2 weeks, which would not be advisable in urgent situations. This controversial issue requires further evaluation.[31]

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Treatment of Gastrinoma

The approach to treatment often depends on the results of preoperative localization studies and findings at exploratory laparotomy. At exploration, 85% of these tumors are found in the gastrinoma triangle with 40% on the surface of the pancreas and 40% outside of the pancreas. Only 15% are found within the substance of the pancreas. Percutaneous transhepatic venous sampling may occasionally provide accurate localization of single sporadic gastrinomas. Resection (enucleation of individual tumors, if technically feasible), and even excision of liver metastases, is associated with long-term cure or disease control.^[1]

Treatment options:

1. Single lesion in the head of the pancreas:^[2-5]
 - Enucleation.
 - Parietal cell vagotomy and cimetidine.

- Total gastrectomy (rarely used with the introduction of current therapies).
2. Single or multiple lesions in the duodenum:[2-5]
 - Pancreatoduodenectomy.
 3. Single lesion in the body/tail of the pancreas:[2-5]
 - Resection of body/tail.
 4. Multiple lesions in the pancreas:[2-5]
 - Resection of body/tail.
 - If residual disease is present, parietal cell vagotomy and cimetidine.
 - Total gastrectomy (rarely used with the introduction of current therapies).
 5. No tumor found:
 - Parietal cell vagotomy and cimetidine.
 - Total gastrectomy (rarely used with the introduction of current therapies).
 6. Liver metastases:[6-13]
 - Liver resection where possible.
 - Radiofrequency ablation or cryosurgical ablation.
 - Chemoembolization of liver.
 7. Metastatic disease or disease refractory to surgery and cimetidine:[14-23]
 - Chemotherapy
 - Somatostatin analogue therapy.

Patients with hepatic-dominant disease and substantial symptoms caused by tumor bulk or hormone-release syndromes may benefit from procedures that reduce hepatic arterial blood flow to metastases (hepatic arterial occlusion with embolization or with chemoembolization). Such treatment may also be combined with systemic chemotherapy in selected patients. Treatment with proton pump inhibitors or H2 blocking agents may aid in control of peptic symptoms.

In the era of proton pump inhibitors and H2 blocking agents, the potentially lethal hyperacidity and hypersecretory states induced by excessive gastrin production can usually be controlled. In a series of 212 patients with Zollinger-Ellison syndrome (ZES), no patients died of causes related to acid hypersecretion. Only 2.3% of those patients had undergone total gastrectomy, and the study cohort had a long median follow-up period of 13.8 years. Although 32% of the patients died during the course of the study, only 50% of the 67 deaths were attributable to

ZES-related causes. Those causes were mainly liver metastases with progressive anorexia and cachexia (67%) or secondary endocrine tumors consequent to multiple endocrine neoplasia type 1 syndrome. The development of bone or liver metastases (especially diffuse liver disease) or of ectopic Cushing syndrome during the study period predicted for decreased survival times. [24]

Current Clinical Trials

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Treatment of Insulinoma

Curative surgical excision, by open laparotomy or laparoscopy, is the treatment of choice when possible. The open surgical approach is used if the tumor is suspected to be malignant because en bloc lymphadenectomy is performed for malignant tumors without distant metastases. Intraoperative ultrasonography aids the localization of tumor extent and the relationship to other anatomical structures.[\[1\]](#)

Treatment options:

1. Single, small lesion in the head or tail of the pancreas:[\[1-4\]](#)
 - Enucleation, if feasible.
2. Large lesion in the head of the pancreas that is not amenable to enucleation:[\[1-4\]](#)
 - Pancreaticoduodenectomy.
3. Single, large lesion in the body/tail:[\[1-4\]](#)
 - Distal pancreatectomy.
4. Multiple lesions (may occur in 10% of patients, often in association with multiple endocrine neoplasia syndrome type 1):[\[1-4\]](#)
 - Distal pancreatectomy with enucleation of tumors in the head of the pancreas.
5. Metastatic lesions in lymph nodes or distant sites:[\[5-12\]](#)
 - Resect when possible.
 - Consider radiofrequency or cryosurgical ablation, if not resectable.
6. Unresectable:[\[13-22\]](#)

- Combination chemotherapy.
- Pharmacological palliation: diazoxide 300 to 500 mg/day.
- Somatostatin analogue therapy.

Patients with hepatic-dominant disease and substantial symptoms caused by tumor bulk or hormone-release syndromes may benefit from procedures that reduce hepatic arterial blood flow to metastases (hepatic arterial occlusion with embolization or with chemoembolization). [6,8-12] Such treatment may also be combined with systemic chemotherapy in selected patients.

Current Clinical Trials

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Treatment of Glucagonoma

As with the other pancreatic neuroendocrine tumors, the mainstay of therapy is surgical resection, and extended survival is possible even when the disease is metastatic. Resection of metastases is also a consideration when feasible.[\[1\]](#)

Treatment options:

1. Single, small lesion in the head or tail of the pancreas:[\[1-4\]](#)
 - Enucleation, if feasible.
2. Large lesion in the head of the pancreas that is not amenable to enucleation:[\[1-4\]](#)
 - Pancreaticoduodenectomy.
3. Single, large lesion in the body/tail:[\[1-4\]](#)
 - Distal pancreatectomy.
4. Multiple lesions:[\[1-4\]](#)
 - Enucleation, if feasible.
 - Resect body and tail otherwise.
5. Metastatic disease in lymph nodes or distant sites:[\[5-12\]](#)
 - Resect when possible.
 - Consider radiofrequency or cryosurgical ablation, if not resectable.
6. Unresectable disease:[\[13-22\]](#)
 - Combination chemotherapy.

- Somatostatin analogue therapy. Necrotizing erythema of glucagonoma may be relieved in 24 hours with somatostatin analogue, with nearly complete disappearance within 1 week.

Patients with hepatic-dominant disease and substantial symptoms caused by tumor bulk or hormone-release syndromes may benefit from procedures that reduce hepatic arterial blood flow to metastases (hepatic arterial occlusion with embolization or with chemoembolization). [6,8-12] Such treatment may also be combined with systemic chemotherapy in selected patients.

Current Clinical Trials

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Treatment of Miscellaneous Islet Cell Tumors

VIPoma

Immediate fluid resuscitation is often necessary to correct the electrolyte and fluid problems that occur as a result of the watery diarrhea, hypokalemia, and achlorhydria that patients experience. Somatostatin analogues are also used to ameliorate the large fluid and electrolyte losses. Once patients are stabilized, excision of the primary tumor and regional nodes is the first line of therapy for clinically localized disease. In the case of locally advanced or metastatic disease, where curative resection is not possible, debulking and removal of gross disease, including metastases, should be considered to alleviate the characteristic manifestations of vasoactive intestinal peptide overproduction.^[1] For more information, see the [Treatment Option Overview for Pancreatic Neuroendocrine Tumors \(Islet Cell Tumors\)](#) section.

Somatostatinoma

Complete excision is the therapy of choice, if technically possible. However, metastases often preclude curative resection, and palliative debulking can be considered to relieve symptoms.^[1] For more information, see the [Treatment Option Overview for Pancreatic Neuroendocrine Tumors \(Islet Cell Tumors\)](#) section.

Other Pancreatic Neuroendocrine Tumors

For these very rare tumors, complete surgical excision is the only curative option when technically possible. Debulking or somatostatin analogues are used for palliation of symptoms if the tumor is functional. For more information, see the [Treatment Option Overview for Pancreatic Neuroendocrine Tumors \(Islet Cell Tumors\)](#) section.

Current Clinical Trials

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Treatment of Recurrent and Progressive Pancreatic Neuroendocrine Tumors

There is no established therapy for pancreatic neuroendocrine tumors that recur or progress after previous therapy.[\[1\]](#) Deciding on further treatment depends on many factors, including:

- The specific cancer.
- Previous treatment.
- Site of recurrence.
- Individual patient considerations.

Attempts at re-resection of local tumors that have recurred or re-resection of metastatic lesions may offer palliation, when technically feasible. Intra-arterial chemotherapy is a consideration for patients with liver metastases. Patients with hepatic-dominant disease and substantial symptoms caused by tumor bulk or hormone-release syndromes may benefit from continuous-infusion intra-arterial chemotherapy or procedures that reduce hepatic arterial blood flow to metastases (hepatic arterial occlusion with embolization or with chemoembolization).[\[2-7\]](#) Such treatment may also be combined with systemic chemotherapy. A variety of systemic agents have shown biological or palliative activity,[\[1,8\]](#) including:

- Somatostatin analogues.
- Radiolabeled somatostatin analogues.[\[9-11\]](#)
- Tyrosine kinase inhibitors (e.g., sunitinib).
- Temozolomide.
- Vascular endothelial growth factor pathway inhibitors.
- Mammalian target of rapamycin inhibitors (e.g., everolimus).

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Latest Updates to This Summary (05/09/2025)

The PDQ cancer information summaries are reviewed regularly and updated as new information becomes available. This section describes the latest changes made to this summary as of the date above.

Editorial changes were made to this summary.

This summary is written and maintained by the [PDQ Adult Treatment Editorial Board](#), which is editorially independent of NCI. The summary reflects an independent review of the literature and does not represent a policy statement of NCI or NIH. More information about summary policies and the role of the PDQ Editorial Boards in maintaining the PDQ summaries can be found on the [About This PDQ Summary](#) and [PDQ® Cancer Information for Health Professionals](#) pages.

About This PDQ Summary

Purpose of This Summary

This PDQ cancer information summary for health professionals provides comprehensive, peer-reviewed, evidence-based information about the treatment of pancreatic neuroendocrine tumors (islet cell tumors). It is intended as a resource to inform and assist clinicians in the care of their patients. It does not provide formal guidelines or recommendations for making health care decisions.

Reviewers and Updates

This summary is reviewed regularly and updated as necessary by the [PDQ Adult Treatment Editorial Board](#), which is editorially independent of the National Cancer Institute (NCI). The summary reflects an independent review of the literature and does not represent a policy statement of NCI or the National Institutes of Health (NIH).

Board members review recently published articles each month to determine whether an article should:

- be discussed at a meeting,
- be cited with text, or
- replace or update an existing article that is already cited.

Changes to the summaries are made through a consensus process in which Board members evaluate the strength of the evidence in the published articles and determine how the article should be included in the summary.

The lead reviewer for Pancreatic Neuroendocrine Tumors (Islet Cell Tumors) Treatment is:

- Jaydira del Rivero, MD (National Cancer Institute)

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Levels of Evidence

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Updated: May 9, 2025

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